Cryptocurrency return predictability and futures-market pressure: a replication and extension of Liu and Tsyvinski (2021)

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Preface

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Abstract

I replicate and extend the cryptocurrency-return predictability framework of Liu and Tsyvinski (2021). Despite differences in sample composition and the unavailability of wallet-user data, the replicated coin market index closely reproduces the original study's main descriptive findings, where cryptocurrency returns exhibit substantial volatility, fat tails, positive skew, and extreme price movements. I also find substantial support for time-series momentum, and Google attention. The network-adoption results remain sensitive to data construction, where individual predictors are noisy and the principal component of network factors positively predicts subsequent returns. In the extension, I investigate whether regulated cryptocurrency futures markets contain incremental information for forecasting short-horizon returns beyond current returns and Google search-based investor attention. Using Bitcoin and Ethereum futures data obtained from the Commodity Futures Trading Commission (CFTC), I construct a Cryptocurrency Futures Pressure Index (CFPI) from the momentum-adjusted open-interest growth, futures basis, and gross asset-manager participation through static and dynamic Principal Component Analysis (PCA). The static CFPI positively predicts subsequent cryptocurrency returns over the one to four week horizons. A stricter dynamic specification, which estimates its inputs using lagged information, retains predictive power primarily over the first two weeks. Among the individual components, momentum-adjusted open-interest growth provides the clearest forecasting signal; whereas the futures-basis is positively associated with future returns, and gross asset manager participation adds little forecasting power. Overall, I find that cryptocurrency futures markets contain short-horizon predictive information beyond established momentum and attention channels, with the strength of the evidence depending on index construction methodology.

Introduction

Cryptocurrency markets offer a unique status quo for studying return predictability as they combine high volatility, rapidly changing investor expectations, and market regimes that shift substantially over time. Liu and Tsyvinski (2021) show that cryptocurrency returns display distinctive empirical patterns that variables such as momentum and investor attention may predict in short horizons. Thus, their framework offers a benchmark for evaluating whether cryptocurrency return predictability can be reproduced with publicly available data and whether additional predictors provide incremental explanatory power beyond the original empirical specifications.

I evaluate and extend the cryptocurrency return-predictability framework developed by Liu and Tsyvinski (2021). Firstly, I replicate the study using publicly available data, examining the descriptive characteristics of cryptocurrency returns, short-horizon time-series momentum, and network adoption-based return forecasting. Although the replication uses a smaller sample of coins in the construction of the coin market and omits wallet-user data due to data-availability constraints, I reproduce the main descriptive patterns and the prominent momentum and investor attention effects discussed in the original study. This replication stage therefore establishes the empirical baseline of the thesis and determines which parts of the original findings remain robust under data constraints.

I then extend the analysis by asking whether futures-market information improves cryptocurrency return forecasting. I motivate this question through the growing role of Bitcoin and Ethereum futures as venues that may capture trading intensity, investor positioning, and futures-based valuation signals. This extension is also motivated by the timing of futures-market development. Bitcoin futures were introduced only near the end of the original study's sample period, Ethereum futures were not yet available, and Liu and Tsyvinski (2021) did not analyze Bitcoin futures as part of their return-predictability framework. In scale, this market has become difficult to treat as negligible. CME Bitcoin futures averaged more than 6,200 contracts per day in July 2018, equal to over 230 million USD in daily notional value, whereas CME's cryptocurrency futures and options suite traded 29.4 million contracts worth more than 1.7 trillion USD in notional value in 2024 according to CME Group (2025). Consequently, using Bitcoin and Ethereum futures data, I construct a Cryptocurrency Futures Pressure Index ($CFPI_t$) from momentum-adjusted open-interest growth, futures basis, and gross asset-manager participation through static and dynamic Principal Component Analysis (PCA) specifications. I therefore evaluate the effectiveness of futures-market pressure on cryptocurrency return forecasting in eight-week horizons.

The thesis makes two main contributions. Firstly, I assess the external reproducibility of the main findings in Liu and Tsyvinski (2021) under transparent data limitations. Secondly, I develop a composite futures-market measure and test whether it contains short-horizon predictive power beyond the established momentum and attention signals. In doing so, I extend the original spot-market return-predictability framework in Liu and Tsyvinski (2021) by adding futures-market information that has become substantially more relevant since the original study's sample period.

1 Article Assessment

1.1 Article Summary

In "Risks and Returns of Cryptocurrency", Liu and Tsyvinski (2021) examine the case study of cryptocurrencies in empirical asset pricing, putting forth the notion that cryptocurrency returns may be primarily driven, and to a certain extent forecastable, by cryptocurrency-specific factors rather than standard macroeconomic control factors. The authors construct a value weighted bundle of cryptocurrency returns labeled the "coin market index" composed of returns from cryptocurrencies with market capitalizations larger than a million U.S. dollars, spanning from 2011 until 2018. The authors later utilize these returns to evaluate theoretical narratives regarding network adoption and mining and production costs, and investigate the phenomena of time series momentum, investor attention and finally crypto valuation ratios in cryptocurrency trading.

The researchers conclude that cryptocurrency returns are extreme compared to those of traditional asset classes, as the coin market index exhibits relatively higher average returns and volatility with a positive skew, large fat tails and substantial daily movements. Additionally, the researchers examine two mediums deemed to potentially be relevant in cryptocurrency pricing: network adoption and production costs. The authors measure network adoption through the proxies of the number of wallet users and active addresses, transaction and payment counts and a principal component of all four primary variables, deducing that returns have a statistically significant positive exposure to network growth. On the other hand, the authors measure production costs through proxies such as electricity pricing, generation and consumption in the U.S. and China, and chipmaker returns. However, the researchers find that there is no substantial evidence to suggest that these factors are being priced in for cryptocurrency returns, with little to no statistical significance.

Furthermore, the authors demonstrate that cryptocurrency returns display strong time series momentum, indicating that current "coin market" returns meaningfully predict return performance over the short-term (one to eight weeks), based on out-of-sample testing. Similarly, the researchers conclude that investor attention, which they measure through the number of Google searches and relevant X posts, has a statistically significant and positive effect on short- term returns; and vice versa for the "negative attention" variable. Similarly, the authors illustrate how proxies for cryptocurrency valuation ratios have relatively lower predictive power, and do not predict future returns effectively.

Finally, the researchers compare the returns of crypto to those of traditional asset classes like equities, commodities and fiat currencies through the usage of equity-factor models and macroeconomic factors. They largely find exposures to be low and not statistically significant, and thus conclude that cryptocurrencies demonstrate proportionately higher alphas because of limited codependency with traditional risk factors. The authors additionally suggest that negative regulatory events coincide with lower consequent returns, and a similar relationship is observed with text-based sentiment analysis derived from Google searches. Lastly, the researchers present evidence against the notion of "beauty contests", stating that a discrepancy between volume and volatility does not add predictive power in forecasting future returns.

1.2 Hypotheses and Development

Liu and Tsyvinski (2021) initially compare the risk-return characteristics of cryptocurrencies and contrast them to traditional assets commonly studied in empirical asset pricing literature. The researchers then introduce cryptocurrency-specific mechanisms and phenomena to forecast future returns. Using a weighted-value cryptocurrency portfolio, the authors investigate structural drivers and return predictability.

Hypothesis 1: Network adoption is a primary determinant of cryptocurrency returns

Findings: Returns have positive exposure on multiple adoption proxies (number of wallet users, active addresses, transaction and payment activity) and on the principal component incorporating all four. The paper concludes that coin market returns "positively predict future adoption growth".

Implications: Network adoption is determined as a central driver to cryptocurrency return dynamics, which is consistent with the narrative of the researchers' finding that expectations of future network adoption are incorporated into today's prices, and that this relationship could potentially be used to forecast future returns.

Hypothesis 2: Production-side factors are not meaningful predictors of cryptocurrency returns

Findings: Utilizing electricity cost proxies based on data from the U.S. and China, the authors conclude that there is no statistically significant relationship between production-side factors and coin market returns.

Implications: While relevant, the study indicates that production costs may not be priced into returns in the coin market, and therefore may not be a central driver of cryptocurrency returns, as implied by their relatively lower explanatory power.

Hypothesis 3: Cryptocurrency returns indicate short term time-series momentum effects

Findings: The authors deduce that coin market returns exhibit statistically significant positive momentum effects, where recent returns meaningfully forecast short term returns.

Implications: The short term momentum effect may also be incorporated into the empirical pricing models for cryptocurrencies.

Hypothesis 4: Traditional asset pricing factors, including valuation ratios, do not add substantial explanatory power in forecasting cryptocurrency returns

Findings: The paper demonstrates that coin market returns largely demonstrate weak and statistically insignificant exposures to traditional equity, fiat currency, and general macroeconomic control variables, and boast systematically high alphas as a result. Similarly, the authors state that cryptocurrency valuation ratios do not add meaningfully statistically significant predictive power.

Implications: General asset pricing factors outlined in previous studies may not be as relevant in the pricing of cryptocurrencies as crypto-specific measures.

Hypothesis 5: Attention and sentiment are effective in predicting cryptocurrency returns

Findings: The researchers find that periods of increased attention, proxied through a Google search measure, predict relatively higher subsequent returns over upcoming short term periods, while negative attention predicts comparatively lower returns. Text-based sentiment also holds statistically significant prediction power, and the authors therefore conclude that cryptocurrency returns are high when investor sentiment is deemed to be high.

Implications: The findings suggest that risk-return dynamics in cryptocurrencies is largely demand and sentiment-driven rather than on pure financial fundamentals. Additionally, the fact that cryptocurrency-specific factors introduced in the study hold higher explanatory power than traditional macroeconomic factors is indicative of cryptocurrency returns' misalignment with commonplace empirical asset pricing norms.

Hypothesis 6: News of regulation and proxies to measure market disagreement are potentially relevant determinants of cryptocurrency returns

Findings: The event-study analyses performed by the researchers suggests that unfavorable (for cryptocurrencies) regulatory news has a negative association with cryptocurrency returns, while the relationship is weak to non-existent for positive news. The authors also study the "beauty contest" notion, the argument that investors buy cryptocurrencies not because they individually demand it but rather because others may do so, and find that cryptocurrency returns are higher "when the current volume-to-volatility ratio is higher", although this measure is not effective at forecasting future returns.

Implications: The findings demonstrate that regulation and market disagreements are potentially relevant measures in the prediction of cryptocurrency returns, especially within case studies with negative news. Additionally, the results indicate that heterogeneous beliefs may be appropriate in modeling cryptocurrency return behaviour, despite not adding predictive power for future forecasting.

1.3 Data, Variables, and Methodological Choices

1.3.1 Data and Key Variables

Liu and Tsyvinski (2021) create a novel dataset consisting of cryptocurrency-native fundamentals such as network adoption and production costs; attention and disagreement as investor sentiment proxies, and finally traditional financial benchmarks and link them to a newly formed "coin market" return variable designating a value-weighted index of the cumulative returns of major cryptocurrencies, in order to test the robustness of traditional empirical asset pricing models as opposed to crypto-specific forces in return prediction.

The "coin market" variable is composed of currencies with market capitalizations larger than a million U.S. Dollars, and the returns of each individual cryptocurrency is derived from the daily closing prices retrieved from CoinMarketCap. These returns are combined with earlier Bitcoin return data (fetched from CoinDesk), and the final coin market index spans from 2011 until 2018. Said returns are analyzed in daily, weekly, and monthly intervals depending on the mechanism being tested.

The crypto-specific variables proposed in the study are network adoption (the number of wallet users retrieved from Blockchain.info; the number of active addresses, transactions and payments retrieved from Coinmetrics, and a principal component including all four), mining costs (U.S. electricity costs, China electricity cost data, Antminer prices), attention (Google searches including the word "Bitcoin" at a higher rate than usual), negative attention ("Bitcoin hack" searches divided by "Bitcoin" searches), sentiment (the log ratio of positive searches to their negative counterparts) and disagreement (the volume-to-volatility ratio). On the other hand, the traditional benchmarks utilized in the study include the Fama-French framework, foreign exchange and commodity data, macroeconomic factors, and stock return data from China and the U.S. obtained from CSMAR and CRSP respectively.

1.3.2 Methodological Considerations and Implementations

Due to the inherent cash-flowless nature of cryptocurrencies, and the implied difficulty of modeling their returns through financial fundamentals, the authors introduce multiple novel proxies per crypto-specific mechanism, and use dimensionality reduction techniques to minimize measurement error and isolate relationships between variables. The paper employs predictive time-series regressions with coin market returns, time series factor exposure regressions, VaR, and event-study linear regressions for regulatory event dates.

In terms of sample selection, the coin market index is composed of 1,707 cryptocurrencies with market capitalizations of above 1 million U.S. Dollars, excluding smaller and presumably more volatile cryptocurrencies which may have mechanically introduced additional idiosyncratic risk or caused spurious regressions. The inclusion of return data strictly post-2011 is motivated

by the notion that pre-2011 Bitcoin trading is limited and thin, and because ascertaining the validity of such data is consequently difficult.

Similarly, Liu and Tsyvinski (2021) transform multiple key variables for increased interpretability and to form comparisons over time. Network adoption and mining costs are expressed as growth rates, attention is defined as abnormal google search density relative to a baseline and standardized, negative attention is introduced to isolate positive attention from general "buzz", sentiment is the log-ratio of positive to negative attention, and finally investor disagreement is expressed through the volume-to-volatility ratio.

Finally, the researchers utilize factor exposure to determine the relationship between cryptocurrency-specific and general market variables and cryptocurrency returns, and predictive linear regressions and portfolio sorts to gauge the effectiveness of momentum, attention, and sentiment on cryptocurrency return forecasting. The t-statistics are adjusted for serial correlation through the usage of Newey-West standard errors. The study's comparison of these models with benchmark sets with traditional macroeconomic factors suggests that the weak traditional factor pricing, and increased predictability exhibited by crypto-specific factors is not model-specific but rather the consensus outcome across various approaches.

1.4 Empirical Method and Significance of the Main Result

In their empirical setup, Liu and Tsyvinski (2021) construct a value-weighted coin market return index which proxies the aggregate cryptocurrency market. This index is made up of 1,707 coins which have market capitalizations higher than 1 Million U.S. Dollars over the years 2011 through 2018. This approach aims to create a "market portfolio" for cryptocurrencies in order to generalize findings across the crypto market while ruling out smaller, lesser reliable and more volatile cryptocurrencies which may distort inference.

The researchers initially run time-series exposure regressions to discern the exposure of traditional empirical asset pricing factors and cryptocurrency-native variables to coin market returns. Two cryptocurrency-specific structural channels are introduced: network adoption, and mining costs. Network adoption is quantified through the number of wallets, active addresses, payments and transactions, and a composite principal component of all four. The regressions for these variables suggest strong statistical significance regarding network adoption and coin market returns. Coin market returns load positively on active address growth, and the R^2 values of each regression are in close proximity to 30%. On the other hand, proxies for mining costs and electricity usage indicate little to no statistical significance between the costs associated with cryptocurrencies and coin market returns.

Additionally, the authors evaluate return predictability using weekly predictive regressions of cumulative future returns (1 through 8 weeks) on current predictors, and group returns by terciles. The researchers use Newey-West standard errors and lags to address autocorrelation stemming from overlapping time horizons. The researchers find a statistically significant

momentum effect, where higher returns in the current week correspond to higher returns in subsequent weeks. Similarly, the high-minus-low tercile spread is approximately 6.9% per week in sample, and approximately 5.6% out of sample. Furthermore, information-derived variables add statistically significant return predictability, where negative attention is associated with lower short-term returns, with one standard deviation increase in negative attention predicting 2% lower returns the subsequent week. Finally, text-derived investor sentiment is deduced to have a statistically significant positive effect on future returns.

Thus, Liu and Tsyvinski (2021) define a cryptocurrency market return index, test the effectiveness of traditional macro-market variables and crypto-native factors using proxies for network adoption, mining costs, attention, and investor disagreement. The paper concludes that cryptocurrency returns are mostly shaped by cryptocurrency-specific factors such as network growth and attention dynamics, rather than through traditional empirical asset pricing risk factors.

2 Article Replication

2.1 Replication Process

In this section, I perform a replication of the results of Liu and Tsyvinski (2021), attempting to reproduce the most central descriptive and predictive findings of the article. Specifically, I emulate the paper's summary statistics and return distributions of the "coin market", cumulative value comparisons, time series momentum and finally the valuation-ratio regressions.

The replication largely follows the original methodology and research design, however, it differs in its method of data collection, as most of the data sources utilized in the study are no longer freely available. Namely, in the original paper, Liu and Tsyvinski (2021) assemble a value-weighted "coin market" index composed of the returns of cryptocurrencies with market capitalizations above 1 Million U.S. Dollars, and splice this index with Bitcoin returns from earlier years, scraping the data from CoinMarketCap (2026). In the present replication, I derive the coin market index from a Kaggle (2026) historical dataset, which features 890 coins in total as opposed to the 1,707 examined in the original study. Additionally, I splice the index with earlier Bitcoin return data encompassing the period between the 1st of January 2011 through the 1st of January 2013, fetched from Investing.com (2026). Therefore, I analyze a smaller cross-section of data than the original study and, as a result, the replication differs in sample coverage. Nevertheless, this setup is designed to preserve the economic structure of the original coin market index with readily available data. Another data constraint is that of the network adoption variables used in the original study. Liu and Tsyvinski (2021) utilize wallet user data from Blockchain.com (2026) and address, transaction and payment data from Coin Metrics (2026) to construct cryptocurrency-native adoption proxies. However, in the present replication, Blockchain.com (2026) data is no longer widely available, therefore I formulate the network adoption proxies solely through Coin Metrics (2026) data.

The main variable utilized throughout the replication process is the coin market index, from which I construct daily, weekly, and monthly aggregate return series for descriptive analysis. Within the replication, I apply the same empirical framework as the original article by building upon the coin market index and examining momentum-based predictability through time series regressions and sort-based testing, and by deducing the effectiveness of cryptocurrency-native valuation proxies in cryptocurrency return forecasting. Due to data availability constraints, an exact replication of the original study is not feasible. Specifically, the cryptocurrency-return dataset originally obtained from CoinMarketCap (2026) is unavailable, and wallet-user data from Blockchain.com (2026) cannot be retrieved. Consequently, I conduct a direct replication

of Liu and Tsyvinski (2021) to the extent permitted by the available data, and highlight the paper's descriptive results, momentum-related phenomena, and valuation-based testing.

2.2 Replicated Results

I successfully replicate the general findings of Liu and Tsyvinski (2021). Table 2.1a demonstrates the summary statistics of the replicated coin market index and individual cryptocurrencies compared to those of the stock market in daily, weekly and monthly frequencies, while Table 2.1b displays the extreme positive and negative daily return characteristics of the newly created coin market index.

Table 2.1

a) Summary statistics

Frequency	Asset	Mean	SD	t-Stat	Sharpe	Skewness	Kurtosis	% > 0	N
Daily	CMKT	0.48%	5.57%	4.63	0.09	0.83	15.64	50.87	2921
Daily	Bitcoin	0.48%	5.56%	4.62	0.09	0.90	15.62	50.43	2921
Daily	Ethereum	0.60%	7.38%	2.86	0.08	0.27	16.05	48.87	1242
Daily	Ripple	0.53%	8.86%	2.66	0.06	6.07	100.56	46.08	1975
Daily	Stock	0.04%	0.92%	1.85	0.04	-0.45	7.95	53.95	2011
Weekly	CMKT	3.47%	16.53%	4.29	0.21	1.71	9.95	56.09	419
Weekly	Bitcoin	3.43%	16.30%	4.31	0.21	1.77	10.38	57.52	419
Weekly	Ethereum	4.78%	24.24%	2.64	0.20	1.47	7.76	51.40	179
Weekly	Ripple	5.68%	45.51%	2.10	0.12	7.82	82.32	46.29	283
Weekly	Stock	0.18%	1.91%	1.93	0.09	-0.47	5.16	58.61	418
Monthly	CMKT	20.58%	70.32%	2.87	0.29	4.42	27.91	58.33	96
Monthly	Bitcoin	19.57%	66.13%	2.90	0.30	4.42	27.35	58.33	96
Monthly	Ethereum	23.27%	65.04%	2.29	0.36	1.47	4.90	48.78	41
Monthly	Ripple	32.69%	137.31%	1.92	0.24	4.10	22.02	38.46	65
Monthly	Stock	0.76%	3.28%	2.28	0.23	-0.37	4.24	66.67	96

b) Extreme daily CMKT return events

Disasters	Counts	%	Miracles	Counts	%
$< -5\%$	255	8.73%	$> 5\%$	316	10.82%
$< -10\%$	83	2.84%	$> 10\%$	121	4.14%
$< -20\%$	16	0.55%	$> 20\%$	26	0.89%
$< -30\%$	3	0.10%	$> 30\%$	11	0.38%

Panel A reports summary statistics for the replicated coin-market index, major cryptocurrencies, and the stock benchmark at daily, weekly, and monthly frequencies. Panel B reports daily CMKT returns below each disaster threshold and above each miracle threshold; percentages use the full daily CMKT sample.

The replicated results exhibit similar general trends as the original paper, specifically the phenomena of extreme volatility, prominent right skew and comparatively higher kurtosis compared to the stock market. At the daily frequency, the coin market and bitcoin indicate mean daily returns of 0.48%, and standard deviations of 5.57% and 5.56% respectively, while the stock market exhibits mean daily returns of 0.04%, and a standard deviation of 0.92%. This considerable disparity in return and risk characteristics between cryptocurrencies and traditional asset classes is consistent with the findings of Liu and Tsyvinski (2021).

This relationship is amplified at lower frequencies; where I find that Ripple and Ethereum display standard deviations of 137.31% and 65.04% respectively at the monthly level, as opposed to a markedly lower value of 3.28% in the case of stocks. These results indicate that during the time period analyzed, cryptocurrencies offered a higher Sharpe ratio than stocks, demonstrating superior risk-adjusted performance despite their considerably higher volatility. Similarly, the summary statistics illustrate that cryptocurrencies express relatively higher kurtosis, and this is further evidenced by Table 2.1b, which suggests that extreme daily returns in either direction occur at considerable rates, with negative extreme returns being eclipsed by extreme positive returns at the daily frequency, indicating that such events may be a driving factor behind the relatively positive return characteristics of the cryptocurrency market.

As Figure 2.1 demonstrates, when the distribution of coin market returns are mapped in both normal and logarithmic form in daily, weekly and monthly frequencies, the return distributions peak around zero, and are accompanied by long tails, indicating intense return abnormality. This is true even in the daily frequency, where the distribution is highly concentrated in the center despite the existence of extreme movements in either direction. This effect is most prominent at the weekly and monthly frequency, where the distributions become wider and more asymmetric, exhibiting clear right-skew, where the extreme positive return abnormalities outweigh their negative counterparts. Thus, Figure 2.1 is in line with the original paper's findings that cryptocurrency returns are highly dominated by extreme downturns, booms, and overall return abnormality.

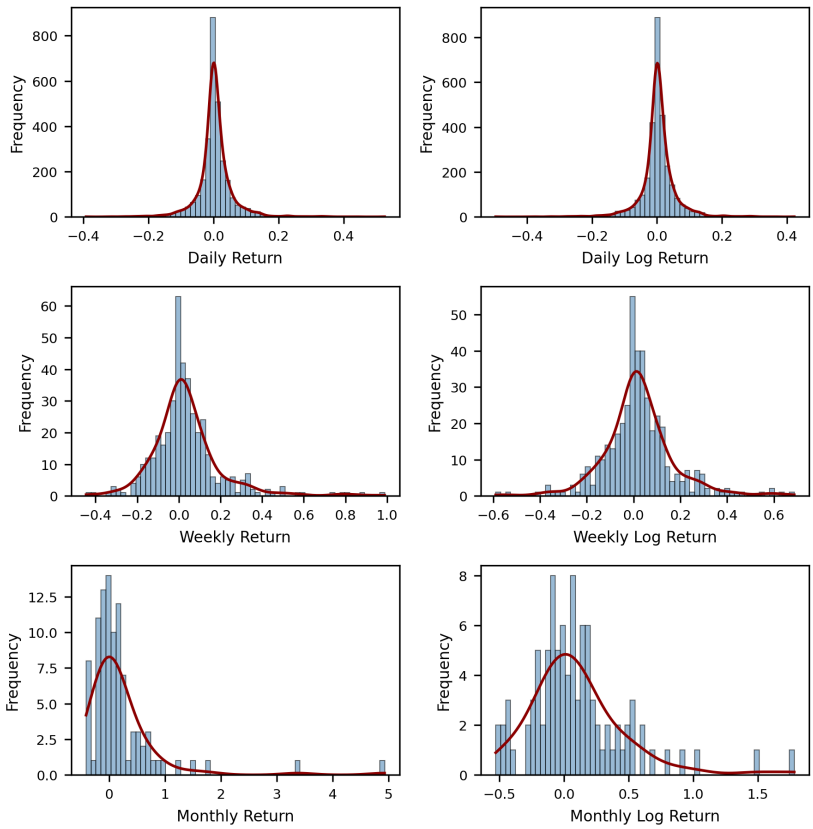


Figure 2.1: Distribution of coin market returns at the daily, weekly, and monthly frequencies.

Figure 2.2 further emphasizes these descriptive findings by visualizing the value of 1 U.S. Dollar invested in the coin market against Bitcoin, Ethereum and Ripple. The figure demonstrates how while Bitcoin closely mirrors the return behaviour of the coin market for much of the sample, Ethereum and Ripple diverge considerably, particularly around 2018. The figure further illustrates that brief periods of high returns are often followed by large reversals, providing visualization consistent with the extreme return characteristics displayed in Table 2.1b. Therefore, the coin market variable's use as a proxy is further motivated by its effectiveness in capturing the aggregate cryptocurrency market dynamics in addition to smoothing out the extremities caused by large fluctuations in price such as those exhibited by Ethereum and Ripple.

Figure 2 — Cryptocurrency market returns and major coins

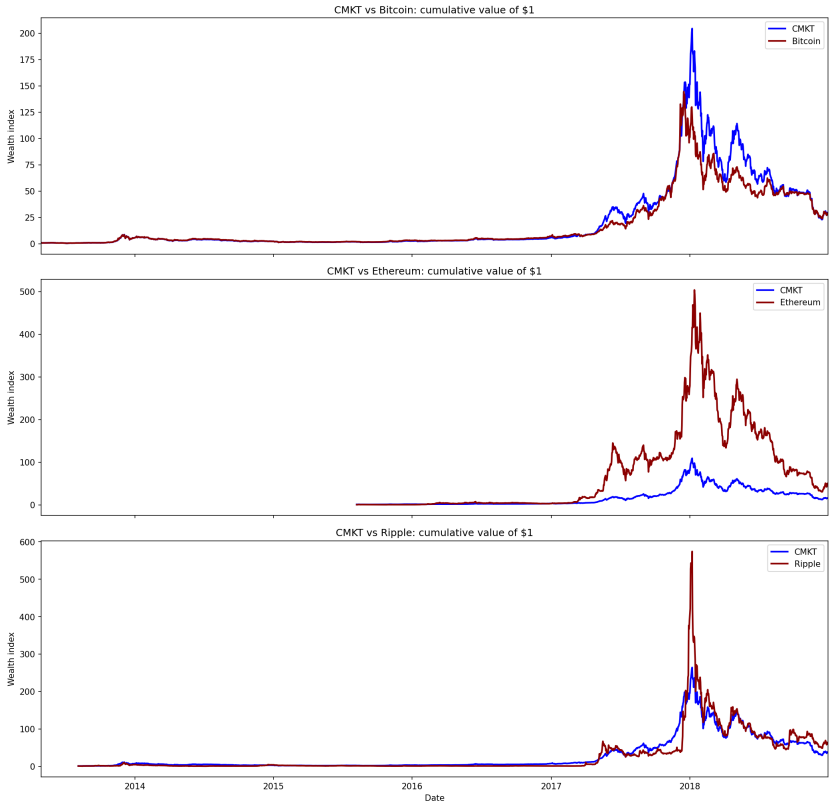


Figure 2.2: Cumulative wealth comparison between CMKT and the major cryptocurrencies.

Table 2.2 directly addresses Liu and Tsyvinski's prediction that cryptocurrency returns exhibit time-series momentum, and the notion that current returns meaningfully and positively forecast future returns. The replication supports this hypothesis in both the regression and portfolio-sort analysis. I find that current weekly coin market returns positively predict future cumulative returns across all horizons, expressed in Table 2.2a, where the coefficient of the momentum variable spans from 0.20 in the one week horizon to 1.56 at the eight-week horizon, increasing monotonically. This effect is statistically significant at the 10% confidence level at the six and eight week horizons, while it indicates statistical significance at the 1% level in the one, two, three and four-week forecasting horizons. Table 2.2b reinforces this pattern, where the high-return tercile earns 7.62% over the next week and 37.12% over the next four weeks, as opposed to 1.37% and 5.83% for the low-return tercile. Additionally, the high-minus-low differences are 6.25%, 10.88%, and 31.29% across each portfolio sort. I further find that the no-lookahead sorting test indicates the same directionality, with differences of 4.38, 8.63, and 25.31 percentage points respectively. Overall, these results support the momentum hypothesis of Liu and Tsyvinski (2021), and demonstrate that cryptocurrency returns display prevalent trend continuation across one-to-eight week time horizons.

Table 2.2

a) Regression results

Weekly	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
R_t	0.20**	0.49***	0.80***	1.07***	1.53*	1.56*
	(2.49)	(2.75)	(3.03)	(2.63)	(1.93)	(1.72)
R^2	0.04	0.08	0.09	0.08	0.05	0.02

b) Sorting results

Time-series momentum by groups (weekly, percentage)

Rank	R_t	$R_{t,t+1}$	t-stat	$R_{t,t+2}$	t-stat	$R_{t,t+4}$	t-stat
Low	-10.61	1.37	(1.14)	3.85	(1.60)	5.83	(1.13)
Middle	1.56	1.41	(1.54)	4.28	(2.28)	13.06	(3.12)
High	19.44	7.62	(4.13)	14.73	(3.73)	37.12	(3.17)
Diff		6.25		10.88		31.29	

Time-series momentum by groups—No lookahead (weekly, percentage)

Rank	R_t	$R_{t,t+1}$	t-stat	$R_{t,t+2}$	t-stat	$R_{t,t+4}$	t-stat
Low	-9.74	0.99	(0.86)	2.60	(1.43)	2.90	(1.00)
Middle	1.78	1.98	(1.86)	3.82	(1.96)	10.32	(2.37)
High	17.09	5.36	(3.01)	11.24	(2.89)	28.21	(2.50)
Diff		4.38		8.63		25.31	

This table reports replicated weekly time-series momentum results. Panel A regresses future cumulative CMKT returns on current CMKT returns; Panel B reports tercile sorts, including a no-lookahead version using first-two-year cutoffs. Newey–West t-statistics use $h-1$ lags. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Lastly, I document that the network adoption-based predictive regression results largely reflect the implications of the network hypothesis proposed in Liu and Tsyvinski (2021), who argue that cryptocurrency-specific adoption measures may help explain and forecast returns due to network value rising with user activity. I find that the valuation proxies proposed in the paper, with the exception of the number of wallets (omitted due to a lack of data availability for replication), exhibit high positive correlation amongst one another, and particularly for the principal component of all included network effects, as discerned from Table 2.3a. For example, Add/MCAP correlates 0.96 with Pay/MCAP and 0.96 with the principal component, while Trans/MCAP correlates 0.97 with the principal component. Such high correlations support the notion that active addresses, transactions, and payments all measure overlapping aspects of cryptocurrency network usage. Table 2.3b signifies that each examined network adoption proxy is individually noisy as a predictor. The sum of the past 100 days of network activity, active addresses divided by market capitalization, the number of transactions divided by market capitalization, and finally the number of payments proportional to the market capitalization

all exhibit negligible and statistically insignificant coefficients across most forecasting horizons. Despite this, the principal component measure composed of all of these variables exhibits a statistically significant positive coefficient in return prediction, with both the magnitude of the coefficient and statistical significance increasing incrementally with widening forecasting horizon. I interpret these results as evidence consistent with the network adoption hypothesis of Liu and Tsyvinski (2021), where network effects are relevant for forecasting cryptocurrency returns, but they are captured more effectively through an aggregate adoption factor rather than through individual network usage proxies.

Table 2.3

a) Correlation of fundamental-to-market ratios

Ratio	Past100	Add/MCAP	Trans/MCAP	Pay/MCAP	PC
Past100	1.00	0.54	0.56	0.53	0.69
Add/MCAP		1.00	0.88	0.96	0.96
Trans/MCAP			1.00	0.91	0.97
Pay/MCAP				1.00	0.96
PC	0.69	0.96	0.97	0.96	1.00

b) Predictive regressions

Predictor	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
Past100	0.00	0.00	0.00	0.00	0.00	0.01
	(0.38)	(0.52)	(0.64)	(0.84)	(1.05)	(1.20)
R^2	0.00	0.00	0.01	0.01	0.02	0.02
Add/MCAP	0.01	0.01	0.01	-0.00	-0.03	-0.08
	(0.71)	(0.38)	(0.11)	(-0.03)	(-0.16)	(-0.25)
R^2	0.00	0.00	0.00	0.00	0.00	0.00
Trans/MCAP	0.03	0.05	0.06	0.08	0.12	0.16
	(1.55)	(1.29)	(0.90)	(0.72)	(0.77)	(0.71)
R^2	0.01	0.01	0.01	0.01	0.01	0.01
Pay/MCAP	0.01	0.01	-0.00	-0.02	-0.03	-0.07
	(0.37)	(0.20)	(-0.02)	(-0.12)	(-0.15)	(-0.23)
R^2	0.00	0.00	0.00	0.00	0.00	0.00
PC	0.01**	0.02**	0.03**	0.04**	0.07**	0.09**
	(2.13)	(2.34)	(2.35)	(2.39)	(2.58)	(2.52)
R^2	0.03	0.04	0.06	0.06	0.08	0.08

Panel A reports Pearson correlations among the replicated fundamental-to-market valuation ratios. Past100 measures recent network activity; Add/MCAP, Trans/MCAP, and Pay/MCAP scale adoption proxies by market capitalization; PC denotes the first principal component. Panel B reports predictive regressions of cumulative future CMKT returns; Newey–West t-statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

2.3 Comparison With Original Findings

The replication is broadly successful in covering and exhibiting the same general trends and conclusions stated in Liu and Tsyvinski (2021), nevertheless, the degree of conformity is relatively higher in the replication of the summary statistics and return characteristics as opposed to the valuation-ratio analysis. I observe the highest degree of correspondence between the replication and the original study in the creation and overall behaviour of the coin market variable. In Liu and Tsyvinski (2021), at the daily frequency, CMKT has a mean return and standard deviation of 0.46% and 5.46%, while in the replication these values correspond to 0.48% and 5.57% respectively. Similarly, the values are nearly equivalent at the weekly frequency, with the original study's mean return and standard deviation figures being 3.44% and 16.50%, and the replication reporting 3.47% and 16.53% respectively. Finally, the replication reproduces the mean and standard deviation of coin market returns as 20.58% and 70.32% at the monthly frequency, while Liu and Tsyvinski (2021) report 20.44% and 70.80%. This coherence suggests that the general trends described in the original study are present in the partial replication, with Bitcoin closely tracking CMKT, Ethereum and Ripple displaying substantial volatility, and cryptocurrencies exhibiting relatively higher risk compared to stocks, yet generating higher returns on average. Thus, I confirm that the replicated CMKT variable remains consistent with the original study, despite a smaller sample of 890 coins compared to 1,707 in Liu and Tsyvinski (2021).

Additionally, the replication closely matches the original paper's findings regarding extreme events and tail behaviour, albeit with higher extremity near the tails. In Liu and Tsyvinski (2021), the researchers find that cryptocurrency returns are positively skewed and exhibit high levels of kurtosis, showing distinct return dynamics inconsistent with stocks. The researchers conclude that both negative and positive extreme daily events occur at "non-trivial rates". The same phenomena are observed in the replication, where daily CMKT returns indicate skewness of 0.83 and 15.64, whereas these values correspond to 0.74 and 15.52 in Liu and Tsyvinski (2021). The results are similar in the weekly and monthly frequencies. Furthermore, extreme event frequencies are largely consistent in the replication, where the reproduced results indicate that 8.73% of daily CMKT return observations are below -5% and 10.82% are above 5%, while Liu and Tsyvinski (2021) find 8.56% and 10.88% respectively. Conversely, I show that the largest divergence is reported in the 10% positive return frequency. I demonstrate that such an event occurs 4.14% of the time whereas the original study indicates a probability of 3.66%. Despite this, the overall narrative of the summary statistics, where cryptocurrencies exhibit extreme fat tails, and that their returns are highly driven by large booms and crashes is consistent with the findings of Liu and Tsyvinski (2021).

The reproduced visualizations, Figure 2.1 and Figure 2.2, closely align with the original study. As described by Liu and Tsyvinski (2021), the coin market return is sharply centered around zero with fat tails, and that the coin market displays co-movement with major cryptocurrencies such as Bitcoin, Ripple, and Ethereum. In the replicated figure, Bitcoin continues to closely match CMKT, whereas Ripple and Ethereum demonstrate greater volatility. This degree of

correspondence carries over to Table 2.2, where the replication supports Liu and Tsyvinski's core time-series momentum finding. In Liu and Tsyvinski (2021), current weekly coin-market returns predict future cumulative returns with coefficients of 0.20, 0.49, 0.81, 1.07, 1.55, and 1.62 from the one to eight week horizons, while Table 2.2 produces nearly identical estimates of 0.20, 0.49, 0.80, 1.07, 1.53, and 1.56, respectively. Likewise, the statistical significance closely mirrors the original paper. Moreover, the portfolio-sort regression results are analogous. In the original paper, the high-minus-low return differences equal 6.91, 12.63, and 31.87 percentage points at the one, two, and four-week horizons; while in the replication, they are marginally smaller but comparable at 6.25, 10.88, and 31.29 percentage points. The no-lookahead results follow the same general trend, with Liu and Tsyvinski reporting differences of 5.62, 11.03, and 29.19 percent, compared with 4.38, 8.63, and 25.31 percentage points in the replication. Overall, I successfully replicate both the direction and magnitude of the original momentum effect, albeit with relatively weaker sorting results, especially for the no-lookahead specification.

Nonetheless, the valuation-ratio analysis yields mixed results, and deviates materially from Liu and Tsyvinski (2021). Firstly, in the original paper, the correlation matrix of the fundamental-to-market ratios indicates that the adoption-based measures are strongly positively correlated with each other, and with the principal component. While this relationship is mostly reflected in the replication, Table 2.3a displays a high in magnitude but comparatively weaker correlation between each variable, especially for "Past100". Additionally, the table remains structurally incomplete due to a lack of access to data for the number of wallets. This limitation causes a disparity between the results of the original study's predictive regressions concerning valuation ratios, and those of the replication. Liu and Tsyvinski (2021) find that individual valuation ratios are consistently negative yet statistically insignificant, and that none of the six valuation ratios meaningfully predict future returns in any time horizon, whereas the replication results suggest that there is no evident pattern of consistently negative coefficients, and similarly report no statistical significance. The most striking discrepancy stems from the principal component of all network measures, where I document that the principal component has a statistically significant positive coefficient which increases with widening time horizon, as denoted in Table 2.3b. This demonstrates that a principal component of network adoption metrics meaningfully predicts CMKT returns. I conclude that this phenomenon indicates an underreaction or caution in the empirical pricing of cryptocurrencies, where increases in underlying network usage may be only gradually capitalized into prices.

2.4 Summary of Replication Effort and Robustness

I replicate the key descriptive patterns identified by Liu and Tsyvinski (2021) and find consistent backing for their predictive models, albeit with distinct exceptions. I document that the strongest aspect of the replication lies with the summary statistics and overall return behaviour of the coin market index, denoted as CMKT. The mean volatility, skewness, kurtosis and extreme positive and negative jumps of the CMKT closely mirror and follow the patterns displayed by the coin market variable constructed in Liu and Tsyvinski (2021), in the daily,

weekly and monthly frequencies. Similarly, the return distribution and cumulative wealth figures reflect the conclusions of the original study; the notion that cryptocurrency returns are exceptionally volatile, extreme-event driven and substantially riskier than those of the stock market. Therefore, despite major differences in sample composition and data availability in its construction, the newly created CMKT variable proves robust, mirroring the conclusions of Liu and Tsyvinski (2021), reinforcing the notion that the index effectively captures the aggregate cryptocurrency market.

The replication outcome of the predictive regressions is comparatively more mixed, but it strongly reproduces the original momentum hypothesis. Liu and Tsyvinski (2021) argue that cryptocurrency returns may exhibit time-series momentum because network externalities and attention-driven trading may cause prices to gradually adjust instead of immediately reflecting the underlying cryptocurrency's current value. I find substantial support for this prediction in Table 2.2. The replicated momentum coefficients closely match the original study, and statistical significance is strikingly similar. Similarly, the portfolio-sorts analysis results in comparable values with the original study across both specifications. Therefore, the replication performs particularly well in reproducing and reinforcing the paper's momentum hypothesis. However, in the network-adoption specifications, the results are mixed. I find that a principal component of relevant adoption measures has a statistically significant positive coefficient which increases with widening time horizon, and that such network adoption proxies are noisy in isolation, consistent with the original study. However, the replication diverges from Liu and Tsyvinski (2021) in finding that the principal component of adoption measures has a statistically significant positive coefficient that increases with the forecasting horizon, suggesting that a collective bundle of network effects may add forecasting power in predicting coin market returns. This may potentially be indicative of a lead-lag relationship, where changes in network effects are not immediately priced in, and experience a persistent under-reaction in which they are corrected in subsequent future returns. However, as the replication omits the independent variable of the number of wallets, which is utilized in the original study, I interpret the results cautiously, as this discrepancy may be due to sample construction and omitted variable bias. Overall, the replication strongly supports the momentum hypothesis proposed in Liu and Tsyvinski (2021), while finding that the network-effect valuation regressions may be more sensitive to data construction.

3 Article Extension

3.1 Updated Literature Review

Liu and Tsyvinski (2021) show that cryptocurrency returns are difficult to explain with standard equity, currency, commodity, or macroeconomic factors, but display predictability through the inclusion of cryptocurrency-specific variables. The researchers identify three crypto-specific patterns: short-horizon momentum, investor-attention and network adoption. In particular, the study finds a strong time-series momentum effect, and that proxies measuring investor attention "strongly forecast future cryptocurrency returns". The researchers start by creating a coin market index that captures the cumulative value-weighted returns of all cryptocurrencies with market capitalizations greater than 1 Million US$. Liu and Tsyvinski (2021) regress future coin market returns on current returns, and find that current coin market returns positively forecast subsequent cumulative coin market returns across all eight horizons, with statistical significance at the 5% level for the first five-week horizons, and at the 10% level for the six-week to eight-week horizons. Similarly, the researchers propose a Google search-based attention proxy, which they conclude to statistically significantly forecast future returns between the one-week to six-week horizons at the 5% level. This coefficient is positive and remains so during the seven-week to eight-week horizons, although it loses statistical significance.

Building on Liu and Tsyvinski (2021), who highlight the predictive power of spot market and attention based variables, I shift the empirical focus to cryptocurrency futures markets, examining whether futures-specific variables such as open interest, futures basis, and trader category participation measures contain additional predictive information beyond the spot market drivers in cryptocurrency return forecasting.

Firstly, open interest refers to the number of outstanding contracts that have not been closed out or delivered at the end of the trading day. Research on conventional futures markets shows that open interest contains significant forecasting power regarding future returns, as it captures traders' sustained exposure to the underlying asset that is not fully incorporated into prices. Hong and Yogo (2012), who examine commodity futures from 1964 until 2008; and currency, bond and stock futures through 1983-2008, indicate that movements in open interest successfully predict future returns in the presence of hedging demand and limited risk absorption capacity in futures markets. The researchers establish that open interest is a better predictor of subsequent returns than futures contract prices, and that this effect is strongest for commodities. Similarly, Bessembinder and Seguin (1993) analyze volatility, trading volume, and market depth across eight futures markets composed of currencies, commodities, and financial

futures. The researchers find that expected future open interest is negatively correlated with volatility across all eight markets examined. They conclude that higher values of open interest are indicative of greater market depth, and subsequently deem high open interest to "mitigate" volatility.

The futures basis, denoted as crypto-carry in Schmeling, Schrimpf, and Todorov (2026), is defined as the difference in price between the futures and spot price of one unit of cryptocurrency. Schmeling, Schrimpf, and Todorov (2026) find that the futures basis is notably large, exceeding 40% per year during certain periods. They argue that such a discrepancy is too large to be explained purely through standard fundamentals such as interest rates and storage costs (which are negligible in cryptocurrencies unlike in commodities), and deem this phenomenon to be an "inconvenience yield" which reflects the cost of holding cryptocurrencies spot relative to futures. In particular, the researchers argue that this relationship is driven primarily by smaller, "trend-chasing" investors creating leveraged long demand through futures, while arbitrage capital is limited by regulatory barriers and the absence of cross-margining due to complex processes required to perform profitable arbitrage. Similarly, Conlon, Corbet, and Oxley (2024) reinforce the notion that the Bitcoin basis exhibits behaviour differing from that of a simple cost of carry residual. Using high-frequency Bitcoin spot and futures data, the researchers find that Bitcoin basis risk is "linked to investor sentiment from established financial markets", particularly through tail risk in various volatility indices such as the VIX. The researchers establish that during periods of extreme volatility, Bitcoin basis is often negative, indicating that Bitcoin futures trade at a lower price than spot. They also determine that this effect is amplified during times of unexpected inflation and deflation.

Cryptocurrency futures data also allows researchers to distinguish broad market activity from the participation of specific trader categories. Baur and Smales (2022) indicate that the launch of regulated Bitcoin futures has given "institutional traders access to a regulated cryptocurrency product". Using CFTC Commitment of Traders data, the researchers conclude that leveraged-money traders, which include hedge funds and commodity trading advisors, hold the largest positions in Bitcoin futures among various trader categories, and play "a central role" in the market. The study reports that leveraged-money traders are typically net short, suggesting that Bitcoin futures are used not only for long exposure but also for speculative short positioning. These findings are particularly relevant for this study as they indicate that Bitcoin futures may potentially be relevant in capturing informed institutional risk-taking, positioning tendencies and speculative demand that may add explanatory power to determining short-run cryptocurrency return dynamics beyond retail attention proxies. On the other hand, Dunbar and Owusu-Amoako (2023) directly examine Bitcoin-futures positioning in forecasting cryptocurrency returns. Using weekly CFTC Commitment of Traders data, they construct a positioning measure for speculative retail traders and find that changes in this measure have a statistically significant effect on returns of Bitcoin, Ethereum, Ripple, and a broader crypto-index closely resembling that of the methodological setup of Liu and Tsyvinski (2021). The researchers find that the effect is statistically significant when controlling for relevant predictors such as Google-search attention, Bitcoin-futures attention, cryptocurrency uncertainty, economic news sentiment, and lagged returns. Furthermore, the study finds that

a one standard deviation increase in net-short speculative positioning increases subsequent returns in the following week by 0.62% for Bitcoin, 0.93% for Ripple, 1.05% for Ethereum, and 1.03% for the broader crypto market.

The literature review therefore identifies a possible research gap on whether open interest, futures basis, and trader category-based signals in regulated cryptocurrency futures markets collectively contain predictive information beyond momentum and investor-attention variables. I examine this question by testing whether a Cryptocurrency Futures Pressure Index (CFPI) predicts short-horizon Bitcoin and Ethereum returns after I control for the previously established momentum and Google search-based attention predictors.

3.2 From Spot-Market Predictors to Futures-Market Signals

This extension examines whether regulated cryptocurrency futures markets contain incremental information for forecasting short-horizon cryptocurrency returns beyond the momentum and investor-attention channels documented by previous literature. Using Commodity Futures Trading Commission (CFTC) data, I construct a Cryptocurrency Futures Pressure Index (CFPI) through the implementation of static and dynamic Principal Component Analysis (PCA), in order to summarize three distinct futures-market signals into a single predictor and evaluate whether it adds forecasting power to the existing cryptocurrency return-predictability framework of Liu and Tsyvinski (2021). Thus, I contribute to the growing body of literature on cryptocurrency return forecasting by consolidating complex cryptocurrency futures data into a composite futures pressure measure and demonstrating that this index captures relevant information not fully captured by spot-market predictors.

3.2.1 Methodology

The analysis follows the predictive-regression logic of Liu and Tsyvinski (2021) but extends the original framework by introducing cryptocurrency futures-market variables. I specifically replicate the "interaction between momentum and attention" regressions from the original study. As Table 4.1 in the appendix demonstrates, I successfully replicate the original regressions, as both momentum and attention show positive coefficients and stay statistically significant mainly at short horizons, while the interaction terms are generally insignificant, suggesting that the two channels indicate individual significance but jointly weak interactions. There are negligible differences between exact coefficients, but the general economic findings of the study persist. I build on these regressions with the addition of the newly constructed Cryptocurrency Futures Pressure Index, $CFPI_t$ as an independent variable.

Thus, the main dependent variable examined in the extension is the cumulative future BTC–ETH cryptocurrency return proxy over horizon h, denoted $R_{t,t+h}$, where $h \in \{1, 2, 3, 4, 6, 8\}$, consistent with the original paper. The explanatory variables I test are current returns, R_t, the

Google-search attention measure, $Google_t$, and finally $CFPI_t$, where the coefficient on $CFPI_t$ captures the incremental predictive content of futures-market pressure after controls.

I analyze data from both Bitcoin and Ethereum to enhance the robustness of my findings and capture broader cryptocurrency market dynamics, rather than isolating Bitcoin-specific implications. Bitcoin is chosen because it is the most extensively studied regulated cryptocurrency derivatives market, while Ethereum adds an additional source of major cryptocurrency derivatives exposure and overlaps with CFTC coverage. I exclude Ripple, as Ripple futures were introduced mid-2025, and there are too few observations to incorporate into the empirical analysis, as CFTC data is reported weekly. Finally, as Bitcoin and Ethereum contracts differ fundamentally in underlying asset, price level and contract size; I convert positions and open interest into USD nominal terms before aggregation. In cases where CFTC data include both standard and micro contracts, I combine these values strictly after notional conversion. I also estimate a Bitcoin-only $CFPI_t$ as a robustness check.

i) Raw and Adjusted Open Interest

Open-interest growth captures changes in outstanding cryptocurrency futures exposure. It is defined as the number of contracts remaining unsettled at the end of the trading day by Hong and Yogo (2012). I use it as a proxy to measure market depth, and traders' willingness to enter into cryptocurrency futures contracts.

I pool Bitcoin and Ethereum open interest in USD notional terms:

$$\mathrm{OI}_t^{Pool} = \mathrm{OI}_t^{BTC} + \mathrm{OI}_t^{ETH}$$

I then calculate weekly open-interest growth from the pooled series:

$$\Delta\mathrm{OI}_t^{Pool} = \log(\mathrm{OI}_t^{Pool}) - \log(\mathrm{OI}_{t-1}^{Pool})$$

Thus, I first sum BTC and ETH open interest in levels. I then calculate the weekly log growth of the pooled series. Through this weighting procedure, I ensure that a Bitcoin and an Ethereum contract are not deemed economically equivalent, and that a larger weight is given to the market with proportionately higher derivatives exposure.

However, this open interest variable indicates a 70.7% correlation with current returns (see table 4.2 in the appendix), raising significant momentum concerns. To address this issue, I residualize pooled open-interest growth with respect to contemporaneous cryptocurrency returns:

$$\Delta\mathrm{OI}_t^{Pool} = \alpha + \beta R_t + \varepsilon_t,$$

and define the adjusted measure as:

$$\widetilde{\Delta \mathrm{OI}}_t^{Pool} = \hat{\varepsilon}_t.$$

This process mechanically removes the component of open-interest growth associated with price movements during the same week. The adjusted open-interest variable has a 0% correlation with same-week returns by design, and has remarkably low correlations with lagged returns from previous weeks. I use adjusted open interest in the main specification, but test raw open interest as a robustness check. I discuss how this adjustment shapes the interpretation of my main findings in the results section.

(ii) The Futures Basis

The second variable captures futures-market pricing pressure through the premium of regulated CME cryptocurrency futures over spot prices. I use Yahoo Finance continuous CME futures tickers, with BTC=F and ETH=F as futures prices and BTC-USD and ETH-USD as the corresponding spot prices.

For each asset $j \in \{BTC, ETH\}$, I define the futures basis as:

$$B_t^j = \ln\left(\frac{F_{cont,t}^j}{S_t^j}\right),$$

where $F_{cont,t}^j$ denotes the continuous CME futures price and S_t^j denotes the corresponding spot price. A positive value indicates that futures trade at a premium to spot, while a negative value indicates a discount.

I align the basis series, which has a daily frequency, to weekly CFTC positioning data by matching the date of CFTC entries with the latest end-of-day basis value available at the time, and choose the most recent available alternative in case of a missing value on an exact date.

I then aggregate the BTC and ETH bases using USD-notional open-interest weights:

$$\omega_t^j = \frac{\mathrm{OI}_t^{j,USD}}{\sum_{k\in\{BTC,ETH\}} \mathrm{OI}_t^{k,USD}}, \qquad \text{and} \qquad B_t^{Crypto} = \sum_{j\in\{BTC,ETH\}} \omega_t^j B_t^j.$$

Thus, the pooled basis assigns more weight to the asset with greater open interest. I interpret B_t^{Crypto} as a CME futures-premium proxy rather than an exact cost-of-carry measure, because the Yahoo Finance futures tickers represent rolled continuous series rather than individual contracts with observable maturity dates.

(iii) Asset Manager Participation

The third variable captures the intensity of asset-manager participation in regulated cryptocurrency futures markets, where I use the asset-manager long and short positions reported in the weekly CFTC data.

For each asset $j \in \{BTC, ETH\}$, I first convert long positions, short positions, and total open interest into USD-notional values. I combine the standard and micro contracts after this step.

I then pool BTC and ETH positions and define gross asset manager participation as:

$$AM_t^{Gross} = \frac{\sum_{j\in\{BTC,ETH\}} \left(AM_{Long,t}^{j} + AM_{Short,t}^{j}\right)}{\sum_{j\in\{BTC,ETH\}} \mathrm{OI}_t^{j}}.$$

Where the numerator captures the total scale of asset-manager exposure. The denominator scales this exposure by pooled BTC and ETH open interest.

This variable measures the degree of participation by asset managers in cryptocurrency futures contracts. A higher value indicates that asset managers account for a larger share of the activity as opposed to other types of traders such as retail or dealers. I interpret this measure as pseudo institutional sentiment, capturing the willingness of professional traders in taking cryptocurrency-specific risk in futures contracts.

3.2.2 CFPI construction

I combine the three futures-market variables into a Cryptocurrency Futures Pressure Index ($CFPI_t$). The index is composed of:

$$X_t \in \left\{ B_t^{Crypto}, \widetilde{\Delta \mathrm{OI}}_t^{Pool}, AM_t^{Gross} \right\},$$

where B_t^{Crypto} measures futures-basis pricing pressure, $\widetilde{\Delta \mathrm{OI}}_t^{Pool}$ captures momentum-adjusted open-interest growth, and AM_t^{Gross} measures gross asset-manager participation. Before the variables are combined into the $CFPI_t$, I apply an inverse hyperbolic sine transformation, as these variables differ substantially in scale and contain extreme observations. This transformation behaves similarly to a logarithmic transformation for large values but remains defined for zero and negative observations, as discussed in Bellemare and Wichman (2020). For each variable X, I calculate:.

$$\psi(X_t) = \operatorname{asinh}\left(\frac{X_t}{\sigma_X + \varepsilon}\right),$$

where σ_X denotes the full-sample standard deviation of X and $\varepsilon = 10^{-8}$ prevents division by zero. I then standardize the transformed variable:

$$z_{X,t} = \frac{\psi(X_t) - \overline{\psi(X)}}{s_{\psi(X)} + \varepsilon}.$$

Standardization ensures that variables with larger variances do not mechanically dominate each principal component. For robustness, I test both a static and a dynamic PCA approach.

3.2.2.1 Static PCA

The main CFPI specification I use is that of the first static principal component of the three standardized variables. Principal Component Analysis (PCA) constructs linear combinations of variables that capture their common variation. The first principal component is the linear combination that explains the greatest percentage of variance, explained in Jolliffe and Cadima (2016). I utilize PCA in this setting as it allows to set weights inside the index relative to the contribution of each futures-market variable in explaining total variance instead of arbitrarily assigning weights. The use of principal components to summarize information from multiple predictors is also consistent with the forecasting approach discussed by Stock and Watson (2002).

Let:

$$\mathbf{z}_t = \begin{bmatrix} z_{B,t} \\ z_{\widetilde{\Delta \mathrm{OI}},t} \\ z_{AM,t} \end{bmatrix}.$$

I estimate the covariance matrix of $\mathbf{z}_t$ over the full extension sample:

$$\widehat{\Sigma}^{Static} = \mathrm{Cov}(\mathbf{z}_t).$$

The static loading vector, $\mathbf{w}^{Static}$, is the eigenvector associated with the largest eigenvalue of $\widehat{\Sigma}^{Static}$. I then define:

$$CFPI_t^{Static} = \mathbf{w}^{Static\prime}\mathbf{z}_t.$$

Finally, I normalize the sign of the PCA loading such that higher index values correspond to stronger futures market pressure and an increase in each individual variable which makes up the index. Otherwise, interpretability weakens due to the possibility of a negative coefficient indicating higher futures market pressure.

The static PCA specification estimates one loading vector over the full sample and applies the same weights to every weekly observation. However, as this method has access to the full sample starting from the first observation, I interpret it descriptively and not as a real time prediction signal.

3.2.2.2 Dynamic PCA

This construction updates the weights of each variable dynamically and iteratively for each period in the sample. It reduces look-ahead bias because each weekly index value is constructed with information that was available prior to that week.

For each variable X, I calculate a time-varying standardized measure using lagged exponentially weighted terms:

$$z_{X,t}^{Dynamic} = \operatorname{asinh}\left(\frac{X_t - \mu_{X,t-1}^{EW}}{\sigma_{X,t-1}^{EW} + \varepsilon}\right).$$

The exponentially weighted mean, $\mu_{X,t-1}^{EW}$, and standard deviation, $\sigma_{X,t-1}^{EW}$, use only observations available before week t. I use a 36-week half-life, which gives larger weight to more recent observations while keeping information from the earlier sample.

I then update the covariance matrix iteratively:

$$\widehat{\Sigma}_t^{EW} = \lambda \widehat{\Sigma}_{t-1}^{EW} + (1-\lambda)\mathbf{z}_{t-1}^{Dynamic}\mathbf{z}_{t-1}^{Dynamic\prime},$$

where:

$$\lambda = \exp\left(\frac{\ln(0.5)}{36}\right).$$

To reduce sensitivity to unstable covariance estimates, I shrink the exponentially weighted covariance matrix toward the identity matrix:

$$\widehat{\Sigma}_t^{Shrink} = (1-\rho)\widehat{\Sigma}_t^{EW} + \rho\mathbf{I},$$

where $\rho = 0.10$. Here, shrinkage improves the stability of covariance estimates by reducing the influence of noisy sampling. This approach follows the regularization notion of Ledoit and Wolf (2004). Finally, the dynamic loading vector, $\mathbf{w}_t^{Dynamic}$, is the eigenvector associated with the largest eigenvalue of $\widehat{\Sigma}_t^{Shrink}$. I define:

$$CFPI_t^{Dynamic} = \mathbf{w}_t^{Dynamic\prime}\mathbf{z}_t^{Dynamic}.$$

Thus, I retain both the static and dynamic PCA specifications for interpretation. I keep static PCA as a simple benchmark, while interpreting dynamic PCA as a stricter no look-ahead bias alternative which updates the loadings of $CFPI_t$ at each iteration based on variable importance during each period.

3.2.2.3 Predictive Regressions

I test whether the CFPI contains incremental information for forecasting cryptocurrency returns beyond the momentum and investor-attention channels documented by Liu and Tsyvinski (2021). For each forecasting horizon, I estimate:

$$R_{t,t+h} = \alpha_h + \beta_h R_t + \gamma_h Google_t + \delta_h CFPI_t + \varepsilon_{t,h},$$

where:

$$h \in \{1, 2, 3, 4, 6, 8\}.$$

The dependent variable, $R_{t,t+h}$, is the cumulative cryptocurrency return over the following h weeks. The current return, R_t, controls for short-horizon momentum. The standardized Google-search variable, $Google_t$, controls for investor attention. The coefficient of the CFPI is δ_h. A statistically significant δ_h indicates that the CFPI contains predictive information beyond the established momentum and attention variables. In order to address potential serial correlation, I report Newey-West standard errors with the maximum lag length set equal to the forecasting horizon, as utilized in Liu and Tsyvinski (2021).

3.2.3 Results and Discussion

I find that futures-market pressure contains incremental short-horizon predictive information, although the strength of the evidence depends on the estimation of PCA weights. In Table 3.1, I show that the momentum-adjusted static PCA specification has a positive and statistically significant coefficient during the time horizon of weeks one through four, which weakens during weeks five through eight but nearly stays within range of statistical significance at the 10% level. The stricter equally weighted shrinkage anchored PCA displays statistical significance at the 5% level in week one, and at the 10% level during week two, losing significance at longer horizons. I observe that the coefficient remains positive throughout the whole forecasting period, similar to the static PCA specification. I attribute this loss of predictive power in the dynamic PCA approach to its formulation, where it uses a more conservative no-look-ahead methodology in which its standardization and weights are estimated strictly using lagged information, unlike the static PCA which uses the full sample to estimate weights a single time at $t = 0$. I further test these specifications by constructing Bitcoin-only adjusted indices, for which I use only Bitcoin-futures based data to formulate each variable instead of the pooled sample, while keeping the dependent variable as pooled BTC/ETH returns. I find that the statistical significance pattern persists, and marginally increases for both specifications, indicating that Bitcoin-futures provide a better proxy for aggregate cryptocurrency futures-market pressure. This finding also suggests that the pooled BTC/ETH futures pressure PCA specifications may add additional noise, which lowers the predictive power of the CFPI.

Table 3.1: Adjusted Open-Interest Benchmark Index Constructions

Variable	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
Panel A: Momentum-adjusted CFPI, static PCA						
Adj. CFPI_t	0.013***	0.016**	0.016**	0.020*	0.020	0.020
	(2.63)	(2.14)	(2.01)	(1.86)	(1.54)	(1.60)
R_t	-0.103*	-0.020	-0.045	0.096	-0.007	-0.012
	(-1.71)	(-0.26)	(-0.48)	(0.99)	(-0.05)	(-0.09)
Google_t	-0.003	-0.009	-0.003	-0.005	-0.011	-0.014
	(-0.31)	(-0.67)	(-0.16)	(-0.25)	(-0.44)	(-0.54)
R^2	0.033	0.024	0.014	0.018	0.011	0.009
Observations	240	239	238	237	235	234
Panel B: Momentum-adjusted CFPI, dynamic PCA						
Dynamic Adj. CFPI_t	0.015**	0.021*	0.019	0.017	0.023	0.025
	(2.11)	(1.91)	(1.26)	(0.90)	(0.81)	(0.63)
R_t	-0.101	-0.025	-0.039	0.142	0.062	0.073
	(-1.61)	(-0.37)	(-0.40)	(1.38)	(0.41)	(0.50)
Google_t	-0.002	-0.006	0.002	0.004	0.000	0.006
	(-0.14)	(-0.41)	(0.14)	(0.20)	(0.01)	(0.27)
R^2	0.028	0.024	0.012	0.014	0.009	0.008
Observations	214	213	212	211	209	208
Panel C: BTC-only adjusted CFPI, static PCA						
BTC Adj. CFPI_t	0.013***	0.019**	0.017**	0.021*	0.022*	0.022*
	(2.58)	(2.35)	(2.11)	(1.94)	(1.67)	(1.81)
R_t	-0.103*	-0.022	-0.046	0.094	-0.009	-0.014
	(-1.72)	(-0.28)	(-0.49)	(0.97)	(-0.07)	(-0.11)
Google_t	-0.003	-0.010	-0.003	-0.005	-0.011	-0.014
	(-0.33)	(-0.72)	(-0.19)	(-0.27)	(-0.46)	(-0.57)
R^2	0.032	0.030	0.015	0.019	0.013	0.011
Observations	240	239	238	237	235	234
Panel D: BTC-only adjusted CFPI, dynamic PCA						
BTC Dynamic Adj. CFPI_t	0.013**	0.018*	0.017	0.018	0.018	0.011
	(1.99)	(1.93)	(1.55)	(1.49)	(1.21)	(0.72)
R_t	-0.080	0.017	0.007	0.183*	0.086	0.083
	(-1.27)	(0.23)	(0.07)	(1.69)	(0.56)	(0.56)
Google_t	-0.002	-0.005	0.004	0.005	0.001	0.006
	(-0.15)	(-0.36)	(0.23)	(0.27)	(0.04)	(0.29)
R^2	0.020	0.016	0.009	0.017	0.006	0.003
Observations	220	219	218	217	215	214

Note: This table compares the main adjusted static PCA index with its dynamic EW shrinkage anchored analogue and with BTC-only robustness indices. Dynamic PCA uses lagged inputs, a 36-week half-life, 10% identity shrinkage, and static-loading sign anchoring. All regressions include R_t and $Google_t$. HAC/Newey–West t-statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

I establish that both the momentum and the attention proxies established in Liu and Tsyvinski (2021) lose most of their statistical significance with the addition of the CFPI, with only momentum boasting a negative statistically significant coefficient in the one week horizon in the static specifications at the 10% level. For robustness, I present the raw open interest-based benchmarks in the appendix. In Table 4.3, I show that it produces stronger results across all specifications, especially for the dynamic approach, although I interpret it as a robustness check due to its high correlation with contemporaneous returns.

Table 3.2 reports the PCA loadings of the variables which make up the static momentum-adjusted CFPI. The futures basis, momentum-adjusted open interest, and asset manager participation display loadings of 0.700, 0.708 and -0.028 respectively. I conclude that futures basis and momentum-adjusted open interest largely drive the static PCA index specification, while asset manager participation makes little contribution to the formulation of the index.

Table 3.2: Static PCA construction of the cryptocurrency futures pressure index.

Variable	Component	Static PC1 weight
Basis_t	Futures-basis pricing pressure	0.700
Momentum-adjusted ΔOI_t	Momentum-adjusted open-interest growth	0.713
Gross AM_t	Gross asset-manager participation	-0.028

Note: This table reports the full-sample static PCA loadings used to construct the main momentum-adjusted cryptocurrency futures pressure index.

However, the interpretation changes meaningfully for the equally weighted shrinkage PCA methodology, where I show that the dynamic CFPI changes composition across different market regimes. I show in Figure 3.1 that adjusted-open interest starts off with a moderate loading, decreases in importance near 2022, and becomes an integral co-driver of the index with the futures basis post-2023, and declines severely afterward. I highlight that the futures basis peaks in terms of loadings during 2024. Gross asset manager participation's loading is strongly negative earlier on in the sample but becomes highly positive during 2024, but rapidly changes its sign twice while retaining its importance in the index, showing high instability.

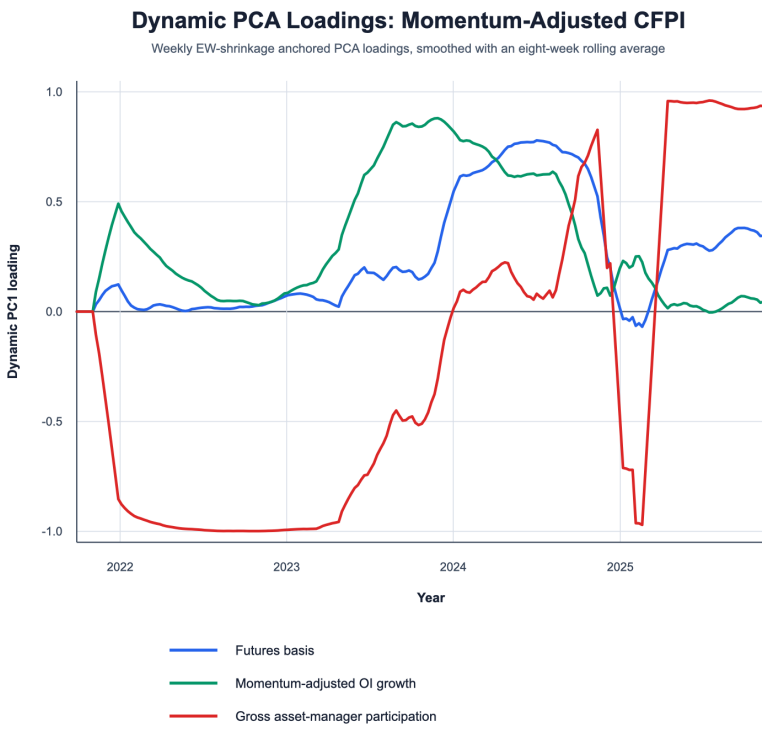


Figure 3.1: Dynamic PCA loadings for the momentum-adjusted CFPI. The signed weekly EW-shrinkage anchored PCA weights are smoothed with an eight-week rolling average.

3.2.3.1 Individual Predictive Regressions

I find that adjusted open-interest growth provides the clearest standalone short-horizon forecasting signal among the CFPI components. Table 3.3 shows the individual predictive regressions for the futures-market variables which make up the CFPI. I document that the coefficient for the futures basis remains uniformly positive, albeit statistically insignificant. Likewise, gross asset manager participation yields a negative and statistically insignificant coefficient across all forecasting horizons.

Table 3.3: Individual predictive regressions for futures-market variables.

Variable	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
Panel A: Futures-basis pricing pressure						
Basis_t	0.682	1.178	0.492	1.179	1.462	1.748
	(1.52)	(1.47)	(0.55)	(0.97)	(0.98)	(1.41)
R_t	-0.100*	-0.018	-0.040	0.100	-0.005	-0.011
	(-1.66)	(-0.22)	(-0.42)	(1.00)	(-0.03)	(-0.08)
Google_t	-0.002	-0.008	-0.001	-0.003	-0.009	-0.012
	(-0.19)	(-0.58)	(-0.04)	(-0.15)	(-0.38)	(-0.50)
R^2	0.017	0.015	0.002	0.009	0.007	0.008
Observations	240	239	238	237	235	234
Panel B: Raw open-interest growth						
Raw ΔOI_t	0.154**	0.161**	0.278***	0.255**	0.203*	0.170
	(2.43)	(2.01)	(2.92)	(2.45)	(1.72)	(1.32)
R_t	-0.248***	-0.171	-0.310**	-0.145	-0.198	-0.170
	(-2.71)	(-1.51)	(-2.44)	(-1.09)	(-1.19)	(-0.96)
Google_t	-0.003	-0.008	-0.004	-0.005	-0.010	-0.013
	(-0.31)	(-0.62)	(-0.22)	(-0.25)	(-0.41)	(-0.51)
R^2	0.032	0.016	0.026	0.019	0.008	0.005
Observations	240	239	238	237	235	234
Panel C: Momentum-adjusted open-interest growth						
Momentum-adjusted ΔOI_t	0.154**	0.161**	0.278***	0.255**	0.203*	0.170
	(2.43)	(2.01)	(2.92)	(2.45)	(1.72)	(1.32)
R_t	-0.098	-0.014	-0.038	0.104	0.000	-0.005
	(-1.63)	(-0.18)	(-0.42)	(1.07)	(0.00)	(-0.04)
Google_t	-0.003	-0.008	-0.004	-0.005	-0.010	-0.013
	(-0.31)	(-0.62)	(-0.22)	(-0.25)	(-0.41)	(-0.51)
R^2	0.032	0.016	0.026	0.019	0.008	0.005
Observations	240	239	238	237	235	234
Panel D: Gross asset-manager participation						
Gross AM_t	-0.048	-0.079	-0.117	-0.144	-0.218	-0.288
	(-1.06)	(-1.02)	(-1.03)	(-1.02)	(-1.16)	(-1.34)
R_t	-0.102*	-0.021	-0.050	0.090	-0.021	-0.036
	(-1.67)	(-0.28)	(-0.53)	(0.89)	(-0.14)	(-0.23)
Google_t	-0.001	-0.007	-0.000	-0.002	-0.008	-0.011
	(-0.14)	(-0.49)	(-0.03)	(-0.11)	(-0.35)	(-0.49)
R^2	0.016	0.012	0.013	0.017	0.022	0.030
Observations	240	239	238	237	235	234

Note: Each panel reports one futures-market predictor with R_t and $Google_t$ controls. Momentum-adjusted ΔOI_t residualizes raw pooled BTC/ETH open-interest growth against contemporaneous R_t. HAC/Newey–West t-statistics use horizon-matched lags. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Most importantly, I find that both raw and momentum-adjusted open-interest growth positively predicts future returns. The coefficients are significant from one to six weeks ahead, with the strongest effect at the three-week horizon. I interpret the stronger significance of momentum in the raw open interest growth regression as mostly mechanical, as the raw variable still contains variation related to momentum. I therefore conclude that the open interest growth's predictive power is not driven primarily by returns, and that it retains statistical significance even when the shared variation with momentum is systematically removed.

In order to measure exactly what adjusted-open interest captures in cryptocurrency return forecasting, I run momentum adjusted open interest on all other independent variables from the extension. Table 3.4 suggests that momentum-adjusted OI growth primarily captures basis-linked futures-market pressure rather than direct asset-manager participation.

I estimate a positive and statistically significant basis coefficient of 1.378 when I include the basis separately, controlling for momentum and the Google-search proxy, and 1.448 in the joint regression with all variables included. Across all specifications, gross asset manager participation has a negative coefficient, with no statistical significance. The joint specification has the highest R^2 value at 6%. Additionally, the near-zero coefficient of the momentum predictor is consistent with the residualization process which adjusted open-interest undergoes, where the effect of current returns is minimized in its construction.

Table 3.4: Determinants of momentum-adjusted open-interest growth.

Variable	Controls (1)	Basis (2)	Gross AM (3)	All (4)
R_t	0.001	-0.004	-0.003	-0.009
	(0.02)	(-0.06)	(-0.05)	(-0.15)
Google_t	0.012*	0.011	0.012*	0.011
	(1.80)	(1.64)	(1.77)	(1.60)
Basis_t		1.378***		1.448***
		(2.99)		(3.09)
Gross AM_t			-0.054	-0.063
			(-1.30)	(-1.58)
R^2	0.019	0.050	0.027	0.060
Observations	241	241	241	241

Note: The dependent variable is momentum-adjusted ΔOI_t, defined as the residual component of pooled BTC/ETH CFTC open-interest growth after removing contemporaneous R_t. The columns report contemporaneous diagnostic regressions rather than causal tests. Column (1) includes the return and Google-search controls; columns (2) and (3) add one revised CFPI component at a time; column (4) includes both components. HAC/Newey–West t-statistics with one lag are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Taken together, these results indicate that the adjusted open-interest is linked most strongly to the futures basis out of all tested variables. However, the relatively low R^2 values across each regression indicate the need for further analysis. I therefore use additional diagnostic checks to

characterize the behavior of adjusted open interest more precisely. Figure 3.2 demonstrates some descriptive statistics for each of the variables which make up the CFPI.

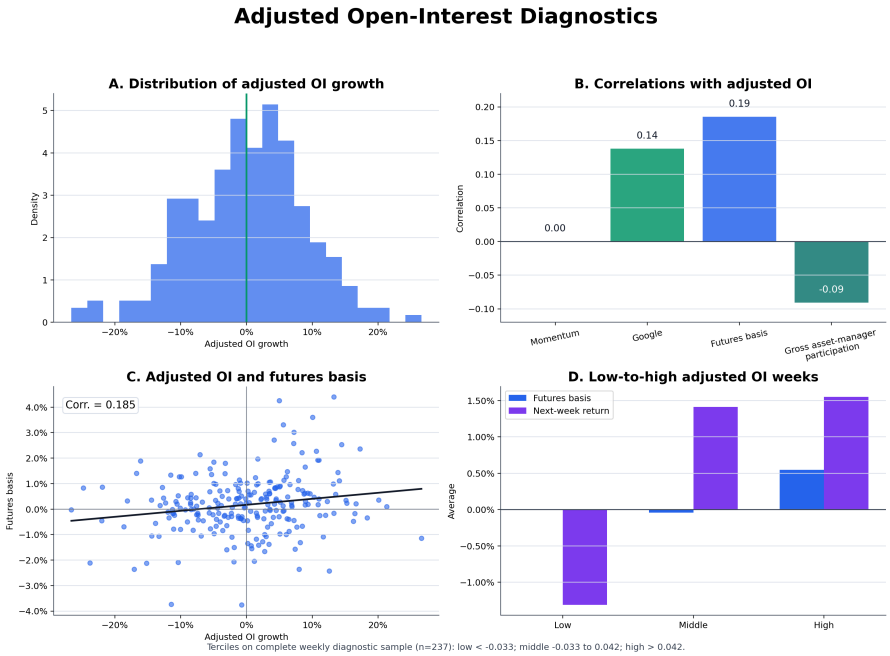


Figure 3.2: Adjusted open-interest diagnostics. Terciles use the 33rd and 67th percentiles.

I show that momentum-adjusted open-interest growth is centered near zero and varies meaningfully across weeks, allowing the CFPI to distinguish between periods of contracting and expanding futures-market pressure. Additionally, adjusted open interest has small positive correlations with Google-attention and the futures basis, at 14 and 19 percent respectively; while it is weakly negatively correlated with gross asset manager participation. I highlight that high-adjusted-OI weeks coincide with a higher average futures basis and stronger next-week returns. This pattern is therefore consistent with the notion that futures-market pressure contains short-horizon predictive information.

In Figure 3.3, I classify weeks into low, middle and high adjusted-OI terciles using the 33rd and 67th percentiles of the complete weekly diagnostic sample. I show that the futures basis increases considerably with each tercile. It starts off slightly negative in the case of weeks with negative adjusted open interest, but increases sharply in the middle and high adjusted open interest tercile weeks. I interpret this futures-basis pattern as speculative demand. When momentum-adjusted open interest grows rapidly, traders may be more willing to pay a premium in order to obtain exposure to cryptocurrency futures contracts. On the other hand, I do not find a monotonic relationship between gross asset manager participation and adjusted open interest. Particularly, the diagnostics indicate that gross asset manager participation is highest in the tercile which includes weeks with low adjusted open interest.

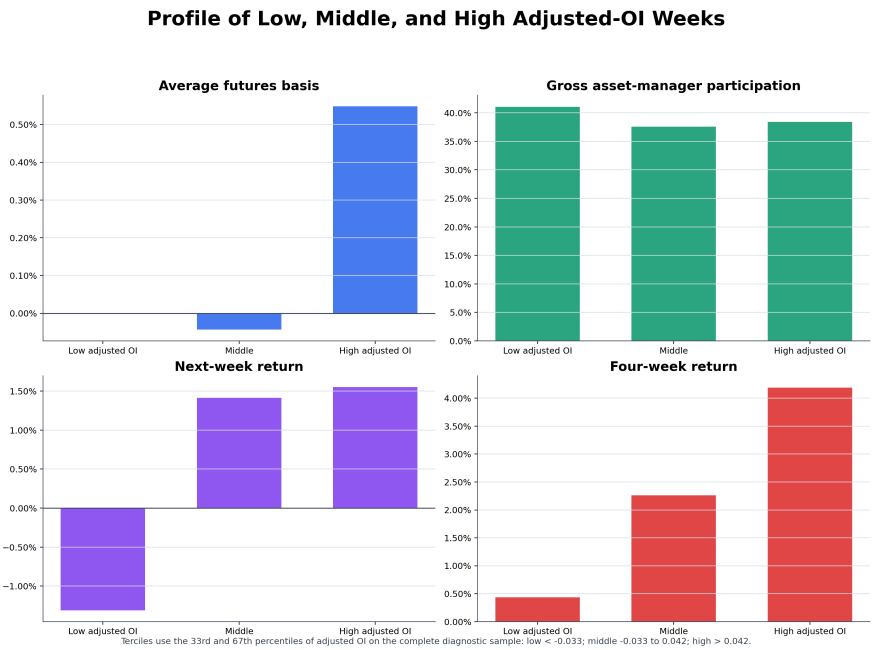


Figure 3.3: Profile of low, middle, and high adjusted open-interest weeks. The panels compare average futures basis, gross asset-manager participation, next-week returns and four-week returns. Terciles use the same 33rd and 67th percentile cutoffs.

I show that average next-week returns rise from approximately -1.3% in low-adjusted-OI weeks to approximately 1.5% in high-adjusted-OI weeks. I find an even stronger effect for four-week returns, where average returns increase from approximately 0.4% in the bottom tercile to more than 4% in high adjusted open-interest weeks. I interpret this pattern as consistent with higher speculative demand being followed by stronger returns, even after removing the variation in open-interest growth associated with same-week momentum. Finally, I conclude that these descriptive statistics resonate with the main findings of the CFPI diagnostics, where stronger speculative futures-market pressure is associated with higher subsequent returns in future periods, particularly within short horizons.

3.3 Outlook on Future Research

Overall, the thesis reverberates the central findings of Liu and Tsyvinski (2021): extreme volatility, fat tails, positive skew, and prominent short-horizon momentum, indicating that the same effects are tangible, and observable despite the usage of distinct data sources. I demonstrate that Bitcoin and Ethereum futures-market variables contain incremental predictive information for short-horizon cryptocurrency returns beyond the cryptocurrency-specific factors examined by Liu and Tsyvinski (2021), although the strength of this evidence depends on index

construction.

Further research may incorporate an updated coin-market variable into the predictive regressions, spanning past 2018 and including delisted coins, updated market capitalizations and data on the number of user wallets, measures which are unavailable in the present replication. The addition of these variables would aid in determining whether the strong statistical significance indicated by the principal component of network variables in the valuation-ratio replication truly exhibits a delayed incorporation of network activity into prices, or whether this phenomenon is sensitive to data construction and the exclusion of the number of wallets. Therefore, future research should expand on the dataset used in the present study, following up with more recent, and complete data regarding both coin market returns and network effects.

Future research should evaluate the out-of-sample performance of the CFPI using longer samples from a wider time period and alternative estimation windows, as the static PCA specification uses full-sample information and the more conservative dynamic specification remains sensitive to calibration choices in terms of statistical significance. Future research may also incorporate futures contracts with known maturities into the construction of the futures basis rather than a continuous Yahoo Finance series in order to isolate expiration-relevant dynamics and produce a more precise futures basis, and include Ripple-based futures contracts for external validity.

Finally, future research may study causal relationships. I interpret this study's analyses as incremental forecasting estimates rather than true causal predictions due to the omission of macroeconomic news, regulatory shocks and other plausible factors which may be of influence to returns and futures positioning. Such variables were not controlled for in the present study because of a lack of data availability. High-frequency event studies that include ETF Flow announcements, futures-expiration dates, or regulatory decisions may help deduce whether the CFPI captures information in short-term cryptocurrency return forecasting, or merely reflects what is already observed in the spot market. Thus, such work would enhance the asset-pricing interpretation of cryptocurrency returns by deducing whether short-term return predictability is driven by a gradual information diffusion, speculative demand or by cryptocurrency-specific risk conditions.

Conclusion

I show that the main empirical patterns documented in Liu and Tsyvinski (2021) concerning coin market behaviour, time-series momentum and the Google-search based attention measure remain robust, even with alternative data collection. Despite the smaller sample of available cryptocurrencies and different data sources, the replicated coin market index displays the same general behaviour as in the original study: high volatility, fat tails, positive skewness, and extreme return characteristics. I also closely reproduce the time-series momentum results in both direction and magnitude, suggesting that short-horizon trend continuation is a persistent feature of cryptocurrency returns. In contrast, I find that the network-adoption results are more sensitive to data construction. The individual adoption measures remain noisy, while the principal component of the available network variables becomes a relatively stronger predictor at longer horizons. This pattern may suggest that the effect of network activity is gradually capitalized into prices, but this finding may be a product of sample construction due to the unavailability of data for the wallet-user predictor from Liu and Tsyvinski (2021) in the replication.

In the extension, I add a futures-market perspective to the return-predictability framework of Liu and Tsyvinski (2021). I find that regulated Bitcoin and Ethereum futures markets contain information that is not fully captured by spot-market momentum or Google search-based attention, particularly during short-horizons. I conclude that the predictive power comes mainly from momentum-adjusted open-interest growth and the futures basis. I interpret these findings as evidence of speculative demand in cryptocurrency futures markets. The diagnostic figures reinforce this interpretation, where weeks with high momentum-adjusted open-interest growth correspond to higher futures-basis values and stronger future returns. Average next-week and four-week returns also increase across the adjusted-open-interest terciles, suggesting that expansions in futures-market pressure are followed by stronger short-horizon cryptocurrency returns even after removing the component of open-interest growth mechanically related to same week momentum. The static CFPI produces stronger predictability across wider time horizons, while the stricter dynamic CFPI's statistical significance wanes past the two week horizon. Taken together, the replication and extension suggest that cryptocurrency prices may adjust gradually to cryptocurrency-specific information, instead of an immediate reflection on the spot market. Future research may explore causal relationships in order to determine whether these patterns are caused by speculative demand, a time delay in information diffusion, or compensation for taking on cryptocurrency-specific risks.

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4 Appendix

Appendix A: Additional Tables

Table 4.1: Interaction between momentum and attention.

Weekly	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
R_t	0.17**	0.43**	0.70***	0.95**	1.42*	1.45
	(2.16)	(2.39)	(2.60)	(2.22)	(1.71)	(1.50)
$Google_t$	0.03***	0.06***	0.08***	0.11***	0.11**	0.11*
	(3.64)	(4.12)	(3.14)	(3.20)	(2.22)	(1.75)
R^2	0.07	0.13	0.12	0.11	0.06	0.02
R_t	0.23**	0.50**	0.85**	1.20*	2.02	2.23
	(2.28)	(2.09)	(2.28)	(1.95)	(1.64)	(1.49)
1Google>0	0.05***	0.10***	0.16**	0.22**	0.34	0.41
	(2.61)	(2.77)	(2.24)	(2.04)	(1.60)	(1.29)
$R_t \times$ 1Google>0	-0.13	-0.15	-0.39	-0.66	-1.69	-2.31
	(-0.77)	(-0.40)	(-0.77)	(-0.91)	(-1.30)	(-1.43)
R^2	0.06	0.10	0.11	0.10	0.08	0.03
$Google_t$	0.03***	0.06***	0.09***	0.09**	0.14**	0.28***
	(2.84)	(2.76)	(4.08)	(2.53)	(2.40)	(2.59)
1R>0	0.04***	0.06**	0.12***	0.17***	0.22*	0.10
	(2.77)	(2.31)	(2.84)	(2.65)	(1.90)	(0.58)
$Google_t \times$ 1R>0	0.00	0.01	0.01	0.05	-0.01	-0.18
	(0.16)	(0.39)	(0.25)	(0.82)	(-0.09)	(-1.44)
R^2	0.06	0.08	0.07	0.07	0.03	0.01

This table reproduces the momentum-attention interaction regressions over the original 2011–2018 sample window. The dependent variable is cumulative future coin-market returns. Newey–West t-statistics are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table 4.2: Return-momentum correlations for raw and adjusted open-interest growth.

Variable	R_t	R_{t-1}	R_{t-2}	R_{t-4}	Raw ΔOI_t	Momentum-adj. ΔOI_t
Panel A: Pooled BTC/ETH CFTC open-interest growth						
R_t	1.000	-0.091	0.078	0.142	0.707	0.000
R_{t-1}	-0.091	1.000	-0.091	-0.020	0.006	0.099
R_{t-2}	0.078	-0.091	1.000	0.078	0.068	0.020
R_{t-4}	0.142	-0.020	0.078	1.000	0.049	-0.072
Raw ΔOI_t	0.707	0.006	0.068	0.049	1.000	0.708
Momentum-adj. ΔOI_t	0.000	0.099	0.020	-0.072	0.708	1.000
Panel B: BTC-only open-interest growth						
R_t	1.000	-0.091	0.078	0.142	0.649	-0.000
R_{t-1}	-0.091	1.000	-0.091	-0.020	0.017	0.100
R_{t-2}	0.078	-0.091	1.000	0.078	0.064	0.019
R_{t-4}	0.142	-0.020	0.078	1.000	0.019	-0.095
BTC raw ΔOI_t	0.649	0.017	0.064	0.019	1.000	0.761
BTC momentum-adj. ΔOI_t	-0.000	0.100	0.019	-0.095	0.761	1.000

This table reports pairwise correlations between current and lagged weekly returns, raw open-interest growth, and the momentum-adjusted open-interest growth variables used in the PCA indices. The momentum-adjusted variable removes the contemporaneous R_t component from raw open-interest growth, so its correlation with R_t should be mechanically close to zero. Lagged-return correlations are reported as a robustness check on remaining return-momentum exposure.

Table 4.3: Raw open-interest benchmark index constructions.

Variable	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
Panel A: Raw open-interest futures pressure index, static PCA						
$CFPI_t$	0.014**	0.019**	0.018*	0.024*	0.026	0.028*
	(2.48)	(2.08)	(1.90)	(1.79)	(1.61)	(1.73)
R_t	-0.182**	-0.127	-0.150	-0.039	-0.158	-0.171
	(-2.54)	(-1.25)	(-1.38)	(-0.34)	(-1.01)	(-1.11)
$Google_t$	-0.003	-0.008	-0.002	-0.004	-0.011	-0.014
	(-0.27)	(-0.64)	(-0.13)	(-0.22)	(-0.43)	(-0.54)
R^2	0.030	0.023	0.013	0.018	0.013	0.012
N	240	239	238	237	235	234
Panel B: Raw open-interest futures pressure index, EW shrinkage anchored PCA						
Dynamic $CFPI_t$	0.019***	0.026***	0.034**	0.039**	0.058**	0.074**
	(2.95)	(2.62)	(2.31)	(2.07)	(2.07)	(2.02)
R_t	-0.165**	-0.113*	-0.158	0.003	-0.148	-0.196
	(-2.43)	(-1.65)	(-1.59)	(0.03)	(-0.86)	(-1.02)
$Google_t$	-0.003	-0.008	-0.000	0.001	-0.005	0.000
	(-0.28)	(-0.58)	(-0.03)	(0.04)	(-0.19)	(0.01)
R^2	0.047	0.043	0.044	0.048	0.059	0.073
N	214	213	212	211	209	208
Panel C: BTC-only raw open-interest futures pressure index, static PCA						
BTC $CFPI_t$	0.013**	0.020**	0.018*	0.022*	0.024	0.025*
	(2.32)	(2.17)	(1.88)	(1.76)	(1.56)	(1.73)
R_t	-0.168**	-0.122	-0.136	-0.019	-0.134	-0.142
	(-2.45)	(-1.19)	(-1.27)	(-0.18)	(-0.91)	(-1.03)
$Google_t$	-0.003	-0.009	-0.002	-0.005	-0.011	-0.014
	(-0.28)	(-0.67)	(-0.15)	(-0.24)	(-0.44)	(-0.55)
R^2	0.028	0.026	0.013	0.017	0.012	0.011
N	240	239	238	237	235	234
Panel D: BTC-only raw open-interest futures pressure index, EW shrinkage anchored PCA						
BTC Dynamic $CFPI_t$	0.017**	0.023**	0.027**	0.031**	0.044***	0.039**
	(2.52)	(2.32)	(2.55)	(2.55)	(2.63)	(2.09)
R_t	-0.160**	-0.089	-0.119	0.038	-0.122	-0.104
	(-2.22)	(-0.99)	(-1.09)	(0.35)	(-0.82)	(-0.69)
$Google_t$	-0.001	-0.005	0.004	0.005	0.001	0.006
	(-0.12)	(-0.33)	(0.24)	(0.28)	(0.02)	(0.26)
R^2	0.026	0.021	0.018	0.027	0.023	0.015
N	220	219	218	217	215	214

This table reports raw open-interest PCA benchmarks. Unlike the preferred adjusted specifications, these indices retain the return-related component of open-interest growth. All regressions control for R_t and $Google_t$. HAC/Newey–West t-statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table 4.4: BTC-only return dependent-variable robustness test: static PCA.

Variable	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
Panel A: Pooled BTC/ETH returns						
BTC Adj. $CFPI_t$	0.013***	0.019**	0.017**	0.021*	0.022*	0.022*
	(2.58)	(2.35)	(2.11)	(1.94)	(1.67)	(1.81)
R_t	-0.103*	-0.022	-0.046	0.094	-0.009	-0.014
	(-1.72)	(-0.28)	(-0.49)	(0.97)	(-0.07)	(-0.11)
$Google_t$	-0.003	-0.010	-0.003	-0.005	-0.011	-0.014
	(-0.33)	(-0.72)	(-0.19)	(-0.27)	(-0.46)	(-0.57)
R^2	0.032	0.030	0.015	0.019	0.013	0.011
N	240	239	238	237	235	234
Panel B: Bitcoin returns only						
BTC Adj. $CFPI_t$	0.013***	0.013*	0.012	0.018*	0.015	0.010
	(2.89)	(1.81)	(1.59)	(1.68)	(1.31)	(0.87)
BTC R_t	-0.100	0.010	-0.049	0.063	0.044	0.028
	(-1.44)	(0.13)	(-0.51)	(0.66)	(0.33)	(0.19)
$Google_t$	-0.003	-0.005	0.001	-0.000	-0.007	-0.009
	(-0.34)	(-0.41)	(0.10)	(-0.03)	(-0.31)	(-0.38)
R^2	0.037	0.017	0.010	0.016	0.007	0.003
N	240	239	238	237	235	233

This table compares the BTC-only momentum-adjusted futures pressure index under the pooled BTC/ETH return outcome and a Bitcoin-only return outcome. Panel A uses cumulative future pooled BTC/ETH returns and includes pooled contemporaneous R_t and $Google_t$ controls. Panel B uses cumulative future Bitcoin returns and replaces the return control with contemporaneous Bitcoin weekly returns. HAC/Newey–West t-statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table 4.5: BTC-only return dependent-variable robustness test: EW shrinkage anchored PCA.

Variable	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
Panel A: Pooled BTC/ETH returns						
BTC Dynamic Adj. $CFPI_t$	0.013**	0.018*	0.017	0.018	0.018	0.011
	(1.99)	(1.93)	(1.55)	(1.49)	(1.21)	(0.72)
R_t	-0.080	0.017	0.007	0.183*	0.086	0.083
	(-1.27)	(0.23)	(0.07)	(1.69)	(0.56)	(0.56)
$Google_t$	-0.002	-0.005	0.004	0.005	0.001	0.006
	(-0.15)	(-0.36)	(0.23)	(0.27)	(0.04)	(0.29)
R^2	0.020	0.016	0.009	0.017	0.006	0.003
N	220	219	218	217	215	214
Panel B: Bitcoin returns only						
BTC Dynamic Adj. $CFPI_t$	0.013**	0.014	0.011	0.013	0.010	-0.004
	(2.12)	(1.56)	(1.23)	(1.16)	(0.78)	(-0.29)
BTC R_t	-0.043	0.031	0.013	0.166	0.144	0.119
	(-0.60)	(0.36)	(0.13)	(1.55)	(0.96)	(0.78)
$Google_t$	0.000	0.001	0.010	0.010	0.004	0.009
	(0.01)	(0.10)	(0.76)	(0.65)	(0.21)	(0.45)
R^2	0.018	0.012	0.011	0.018	0.006	0.003
N	220	219	218	217	215	213

This table repeats the dependent-variable comparison for the no-lookahead EW shrinkage anchored PCA version of the BTC-only momentum-adjusted futures pressure index. The Bitcoin-only specification residualizes BTC open-interest growth with respect to contemporaneous Bitcoin weekly returns before PCA construction. HAC/Newey–West t-statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Appendix B: AI usage and Disclosure Policy

In accordance with the RSM MSc Thesis Manual GenAI Policy, I, Berk Yılmaz, hereby declare usage of OpenAI's Codex, and Grammarly for occasional assistance.

- **Codex**: I made use of ChatGPT 5.5 for reviewing coding pipelines and troubleshooting when writing code in Python.
- **Grammarly**: I used Grammarly for making grammatical corrections, and fixing punctuation mistakes.

Prompts I used within Codex were variations of:

1. Refine my Python code in section X so it is cleaner and easier to follow by strictly adhering to the methodology
2. Check whether I have made any coding errors while implementing my methodology
3. Help me reorganize this notebook cell so that the output renders in the intended place
4. Ensure that I have made no spelling errors in the captions of my tables and figures in section X
5. Review this plotting/export code for syntax issues

All ideas, methodology, analysis, and interpretation are my own.

Disclosure Policy

I, Berk Yılmaz, agree that the data and code submitted with this thesis may be uploaded to a EUR open-access repository and made available through Quarto.

Appendix C: Acknowledgements

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